



ME 324 Design of Thermal Systems

Final Project – Spring 2026

Natural Convection And Heat Sink Design

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1 Part 1: Natural Convection Between Parallel Vertical Plates

1.1 Objective and Problem Description

Part 1 of this project investigates steady natural convection between two vertical parallel plates and validates the simulation results against the analytical correlation of Bar-Cohen and Rohsenow [1]. The correlation predicts the channel-averaged Nusselt number as a function of gap width, plate height, and air properties, covering the full range from the fully-developed (narrow gap) to the isolated-plate (wide gap) limit:

$$Nu_S = \left[\frac{576}{\left(Ra_S \cdot \frac{S}{L}\right)^2} + \frac{2.873}{\left(Ra_S \cdot \frac{S}{L}\right)^{0.5}} \right]^{-0.5} \quad (1)$$

Where $Nu_S = hS/k$ is the gap-based Nusselt number and Ra_S is the Rayleigh number based on gap width S .

Two plate spacings are simulated. The **narrow case** ($S = 3$ mm, $Ra_S = 84.75$) sits in the fully-developed regime where boundary layers from each plate merge across the gap. The **wide case** ($S = 12$ mm, $Ra_S = 5424$) sits near the isolated-plate limit where each plate acts independently. This part focuses on the narrow case first; the wide case result is included in the final comparison.

1.2 Simulation Setup

Geometry

The Icepak domain is a $70 \times 200 \times 100$ mm cabinet. Both plates are 1 mm thick and 80 mm tall, spanning $y = 0.04$ m to $y = 0.12$ m, and are centered around $x = 35$ mm with separation distance S .

Boundary Conditions

Both plate surfaces are held at a fixed temperature of $T_w = 60$ °C. Ambient air enters at $T_\infty = 20$ °C; the temperature difference driving convection is $\Delta T = 40$ °C. Cabinet walls perpendicular to the plates are open (free inlet/outlet); all other walls are adiabatic.

Fluid Properties and Rayleigh Number

Air properties are evaluated at the film temperature $T_f = 40$ °C:

$$k = 0.02662 \frac{\text{W}}{\text{m}\cdot\text{K}}, \quad \nu = 1.702 \times 10^{-5} \frac{\text{m}^2}{\text{s}}, \quad Pr = 0.7255, \quad \beta = \frac{1}{313} \text{ K}^{-1}$$

The Rayleigh number based on gap width is:

$$Ra_S = \frac{g\beta\Delta T S^3}{\nu^2} \times Pr = 3.139 \times 10^9 \cdot S^3$$

$$Ra_S = \begin{cases} 84.75 & S = 3 \text{ mm (narrow)} \\ 5424 & S = 12 \text{ mm (wide)} \end{cases}$$

The solver uses the Boussinesq approximation for buoyancy. All simulations are steady-state.

1.3 Initial Simulation Results

Figure 1 shows the temperature contour from the first run (Run 00, 10 cells across the gap). The chimney effect is clearly visible: cool air is drawn in at the bottom of the channel, heats up along the plate surfaces, and exits as a hot plume at the top. The temperature distribution is symmetric about the channel centreline, and the boundary layers on each plate merge near mid-gap, consistent with the fully-developed behaviour expected at $Ra_S = 84.75$.

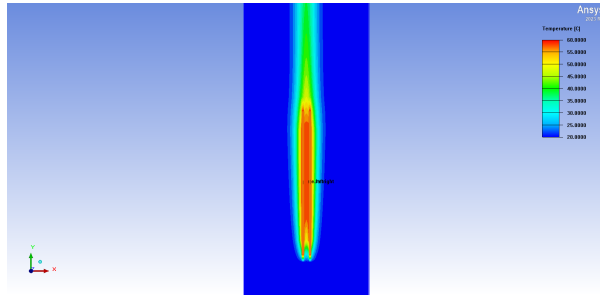


Figure 1: Run 00 temperature contour, narrow case ($S = 3$ mm).

Figure 2 shows the convergence history. All residuals fall below 10^{-3} within 100 iterations with no oscillation, indicating a stable solution.

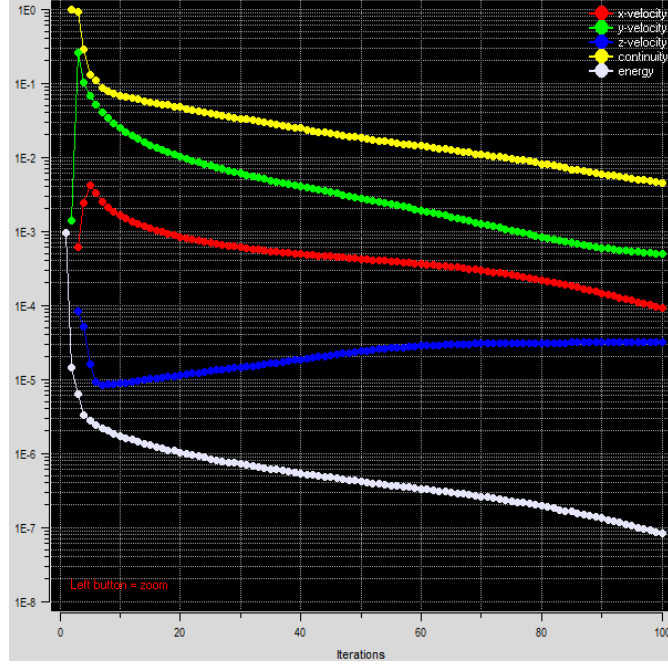


Figure 2: Residual history for Run 00.

Qualitatively the result looks correct, but the computed $Nu_S = 0.1598$ sits +22.4% above the B&R prediction of 0.1306. The setup is clearly capturing the right physics, but the level of disagreement is too large to accept. The most obvious candidate is mesh coarseness across the gap, so the next step is to systematically refine the X -direction resolution and see how the result changes.

1.4 Mesh Refinement

Starting from Run 00, the number of cells across the 3 mm gap was increased in steps – 20, 40, then 60 – while everything else was left unchanged. Table 1 shows the results.

Table 1: Narrow case: X -direction refinement. B&R prediction: $Nu_S = 0.1306$.

Run	Gap cells	Elements	Nu_S (sim)	Sim vs B&R
00	10	128,960	0.1598	+22.4%
01	20	~260k	0.1426	+9.2%
03	40	517,760	0.1486	+13.8%
04	60	777,000	0.1503	+15.1%

Going from 10 to 20 cells cuts the error roughly in half, which is encouraging. But then refining further to 40 and then 60 cells makes the result worse, not better.

The X -direction refinement alone is clearly not converging to the right answer, and a different source of error must be at play.

1.5 Mesh Anisotropy

Looking at the mesh more carefully, it struck me that the cells were quite anisotropic. The gap direction (X) was being refined with each run, but the Y -direction cells – running along the 80 mm plate height – were left at the Icepak baseline throughout. In that direction each cell spanned roughly 6.7 mm, meaning the entire development of the flow from inlet to outlet was captured by only around a dozen cells. For natural convection this matters: the buoyancy-driven velocity builds from zero at the bottom, the boundary layer thickens continuously toward the top, and the air temperature rises along the full plate height. A coarse Y mesh smears all of that through numerical diffusion, mixing air that should be stratified and artificially inflating the predicted h .

To test this, the Y resolution was increased by a factor of 10 on Run 01 – the 20-gap-cell case – since that was the best result so far. Table 2 shows the before and after.

Table 2: Effect of Y -direction refinement, narrow case. B&R: $Nu_S = 0.1306$.

	Gap cells	Y res.	Nu_S (sim)	Δ vs B&R
Run 01	20	Baseline	0.1426	+9.2%
Run 05	20	10 $\times$ fine	0.1330	+1.9%

The error drops from +9.2% to +1.9%, confirming the diagnosis. The coarse Y mesh was responsible for most of the disagreement with B&R all along, and the X -only refinement was chasing the wrong thing. The remaining +1.9% is physically justifiable – B&R assumes an infinite ideal channel, while the Icepak cabinet has finite open boundaries that introduce mild entrance and exit effects.

With this confirmed for the narrow case, the next step is to apply the same approach to the wide case.

1.6 Applying the Findings to the Wide Case

The wide case ($S = 12$ mm, $Ra_S = 5424$) was simulated following the same progression. The first run used 10 gap cells and the baseline Y mesh, giving $Nu_S = 3.418$ against a B&R prediction of 3.137 – a +8.9% disagreement, very similar in character to the narrow case.

The mesh was then set to 20 gap cells with a 10 $\times$ finer Y resolution. Table 3 summarises both runs.

Table 3: Wide case mesh study. B&R prediction: $Nu_S = 3.137$.

	Gap cells	Y res.	Nu_S (sim)	Δ vs B&R
Naive run	10	Baseline	3.418	+8.9%
Fine run	20	10 $\times$ fine	3.161	+0.8%

The improvement mirrors the narrow case exactly. The wide case best run lands at +0.8%, even closer to B&R, which is consistent with the wide case being less sensitive to entrance effects. Figure 3 shows the temperature contour for the fine run. In contrast to the narrow case, the boundary layers on each plate remain thin and well-separated across the full channel height, characteristic of the isolated-plate regime at high Ra_S .

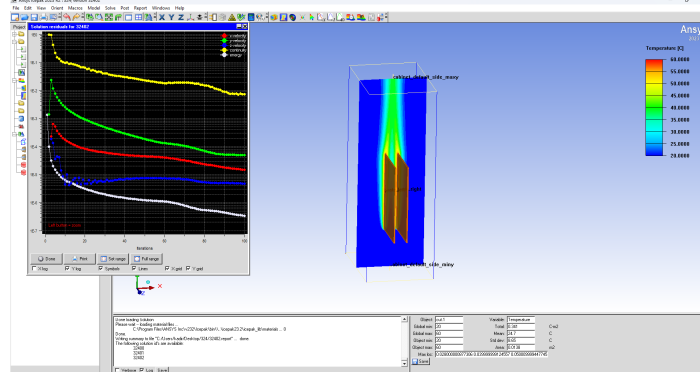


Figure 3: Wide case fine run ($S = 12$ mm): temperature contour.

1.7 Final Comparison with Bar-Cohen & Rohsenow

With both cases validated, Run 05 ($Nu_S = 0.1330$) and the wide fine run ($Nu_S = 3.161$) are placed on a log-log plot against the full B&R correlation curve. Figure 4 shows the result.

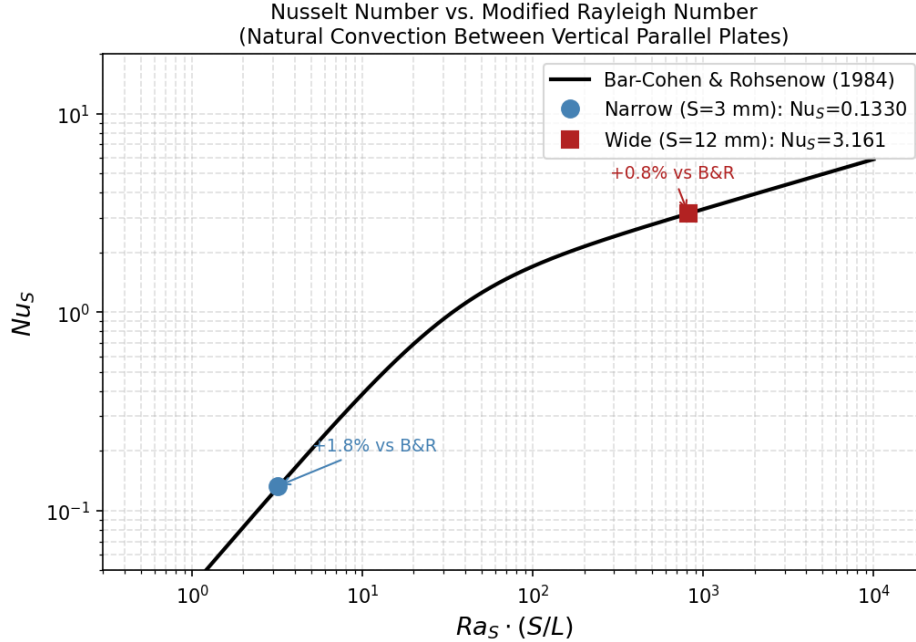


Figure 4: Nu_S vs. $Ra_S \cdot (S/L)$. Solid line: B&R (1984); markers: Icepak best runs.

The horizontal axis uses the similarity parameter $Ra_S \cdot (S/L)$, which collapses both asymptotic limits onto a single curve. The narrow case sits in the fully-developed region on the left and the wide case near the isolated-plate region on the right. Both land close to but slightly above the B&R line, consistent with the small positive bias from the finite

cabinet discussed earlier.

The agreement across both regimes gives confidence that the mesh strategy and solver settings are correctly capturing the buoyancy-driven physics, and provides a validated baseline for the heat sink optimisation in Part 2.

1.8 Icepak Workflow: Frictions and Findings

This subsection documents the practical difficulties encountered during the Icepak setup and what was learned from them. It is not part of the formal analysis but is included because troubleshooting these issues took a significant share of the total time and the solutions are non-obvious to a first-time user.

Wall Objects vs. Solid Blocks

The first attempt modelled the plates as Icepak *wall* objects – zero-thickness surface BCs placed at the plate coordinates. Wall objects can apply a temperature or heat flux condition but have no physical volume or thickness. This became a problem when trying to assign realistic plate geometry: a wall has no mass, no conductivity, and no thickness to carry over into the heat sink design in Part 2. The plates were rebuilt as hollow *block* objects with 1 mm wall thickness, which behave as physical geometry and support the same fixed-temperature BC while being directly extensible to fin and heat sink bodies later.

Domain Sizing

The first attempt used a cabinet that was too tightly sized around the plates. Icepak needs room outside the channel for the thermal boundary layer to develop and for the inlet/outlet flow to settle. The cabinet was revised to $70 \times 200 \times 100$ mm with extra clearance above and to the sides of the plates. Undersized domains can artificially confine the buoyant plume and distort the inlet conditions.

Mesh Not Regenerating After Parameter Changes

After changing the number of gap cells, Icepak sometimes reported “Nothing has changed” and reused the old mesh. The fix was to explicitly press the **Load** button in the Mesh Control dialog rather than relying on automatic regeneration. This is easy to miss and can silently invalidate a run.

Solution ID Management

Icepak names solution saves using the project name plus an incrementing number. When setting up a restart run, the Solve dialog pre-fills the Solution ID field with the last used ID. On one occasion this was not changed before solving, which would have overwritten

an earlier run. The solve was interrupted in time, but the lesson is to always verify the Solution ID field before hitting Solve.

Warm-Starting on Fine Meshes

Fine meshes ($\sim 1.85\text{M}$ elements) diverged when started from a zero-velocity initial condition. The buoyancy force drives an initial surge that the solver cannot damp, leading to oscillating residuals that never settle. Loading a converged coarse-mesh solution as the initial condition via **Post** $\rightarrow$ **Load Solution** eliminates this: the flow field is already physically reasonable and the fine mesh only needs to refine it.

Extracting the Heat Transfer Coefficient

The surface-averaged h is not displayed by default. The correct path is **Report** $\rightarrow$ **Summary Report** $\rightarrow$ select the plate face $\rightarrow$ set quantity to *Heat Transfer Coefficient*, direction to *Max X*, statistic to *Mean*. Initially all faces of the block were selected including the outer surface, which returned an incorrect h . The face filter in the Summary Report panel restricts the selection to the inner channel-facing surface only.

Mass Flow Rate

The channel mass flow rate cannot be extracted from the plate objects. It must be read from a cabinet face: **Summary Report** $\rightarrow$ `cabinet_default_side_miny` (bottom opening) $\rightarrow$ *Mass Flow Rate*. This gave $\dot{m} = 3.249 \times 10^{-4}$ kg/s for the narrow best-case run, implying a mean channel velocity of approximately 0.96 m/s.

Solver and System Notes

Windows Firewall flagged several Icepak solver executables on first launch and blocked them. Each had to be manually allowed before the solver would run. The solver supports parallel execution; the parallelisation settings are accessible in the Solve dialog and were used to speed up the finer mesh runs.

2 Part 2: Heat Sink Design

2.1 Objective and Problem Description

Part 2 of this project focuses on designing a heat sink for a chip dissipating $\dot{Q} = 10$ W inside a naturally ventilated cabinet. The cabinet has no fans; cooling is driven entirely by buoyancy – warm air rises through the cabinet and exits through an outlet grille. The objective is to select heat sink geometry – baseplate dimensions, base thickness, number of fins, fin height and fin thickness – such that the chip temperature is kept as low as reasonably achievable with a modest design.

The chip occupies a 5×5 cm footprint on the top surface of the PCB and sits at the centre of the cabinet. The heat sink sits directly on the chip. All cabinet walls are adiabatic and radiation is neglected, so the only heat removal path is convection from the fin surfaces to the buoyancy-driven air stream.

The design process follows three steps. First, a separate Icepak sub-simulation is used to characterise the heat spreading resistance R_{spread} of the baseplate and select appropriate base dimensions. Second, the Bar-Cohen & Rohsenow correlation provides an analytical starting estimate for the optimal fin spacing and number of fins. Third, the full cabinet model – chip, PCB, heat sink, inlet and outlet grilles – is simulated in Icepak and the design is refined through a parametric sweep over fin number and base geometry.

2.2 Simulation Geometry and Boundary Conditions

Cabinet model built in Icepak 2023 R2: rectangular enclosure, adiabatic walls, opening boundaries at inlet and top-centre outlet grille. The top-centre outlet was selected over a back-wall outlet after a preliminary comparison – it allows the buoyancy plume to exit without recirculation. Chip: 5×5 cm, $\dot{Q} = 10$ W, mounted on PCB 3 mm above floor. Flow: laminar, steady, Boussinesq natural convection, gravity $-y$, $T_{\infty} = 20$ °C, no radiation.

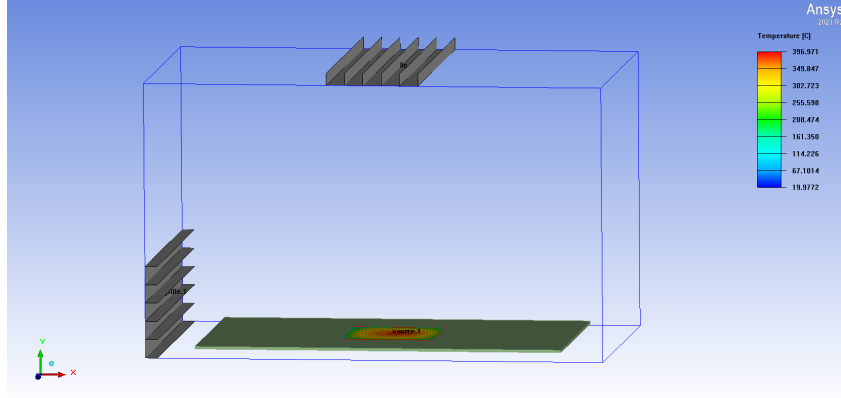


Figure 5: Baseline temperature contour (no heat sink). Natural convection from the bare chip is insufficient; chip temperature exceeds 400 °C.

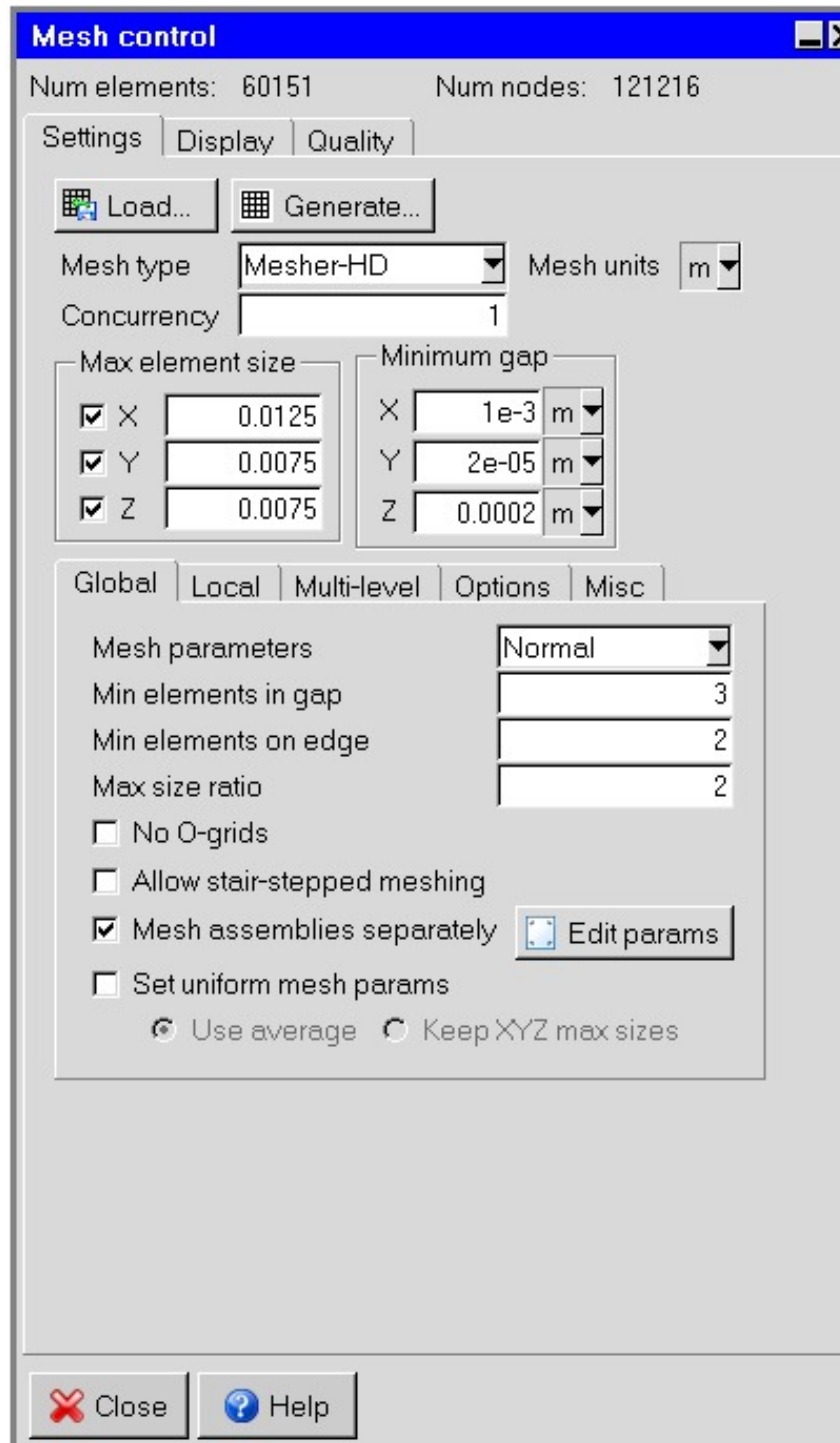


Figure 6: Icepak mesh for the full cabinet model. Mesh is refined around the chip and fin surfaces where temperature gradients are highest.

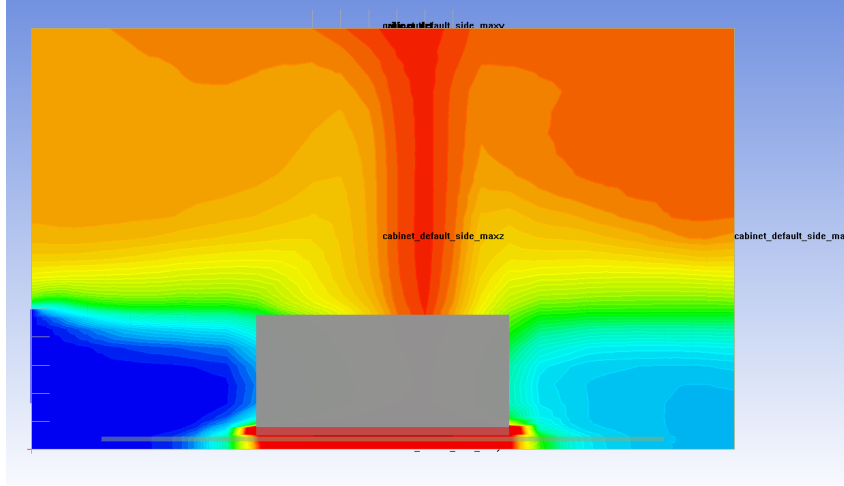


Figure 7: Buoyancy-driven plume from the heat sink after iteration 1. The top-centre outlet arrangement is visible.

2.3 Spreading Resistance Characterisation

R_{spread} is defined as

$$R_{\text{spread}} = \frac{\bar{T}_{\text{chip}} - \bar{T}_{\text{base,top}}}{\dot{Q}}$$

A $3 \times 7 = 21$ -case parametric sweep was run over footprints (6×6 , 8×8 , 10×10 cm) and base thicknesses (2–8 mm). Results are in Table 4 and Figure 8.

Table 4: Chip temperature and spreading resistance. $T_{\infty} = 20$ °C, $\dot{Q} = 10$ W.

Footprint	Thickness (mm)	T_{chip} (°C)	R_{spread} (K/W)
6×6 cm	2	297.9	0.015
	4	297.9	0.012
	6	208.8	0.011
	8	138.6	0.011
8×8 cm	2	176.9	0.069
	4	170.0	0.038
	6	94.8	0.027
	8	75.2	0.022
10×10 cm	2	121.3	0.127
	4	117.4	0.066
	6	98.2	0.046
	8	82.7	0.038

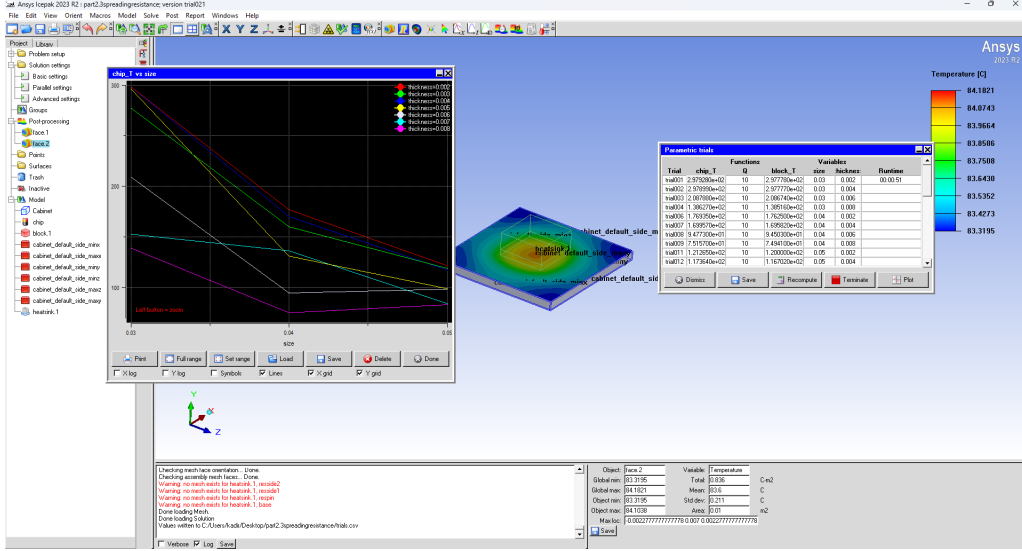


Figure 8: Spreading resistance study: T_{chip} and R_{spread} vs. base thickness for three footprint sizes.

R_{spread} is 0.01–0.13 K/W across all cases – negligible vs. total resistance of 5–28 K/W. Footprint area dominates; base thickness has diminishing returns beyond 6 mm. Footprint 8×8 cm selected for subsequent optimisation.

2.4 Heat Sink Design and Optimisation

2.4.1 Analytical Starting Point: Bar-Cohen & Rohsenow

The Bar-Cohen & Rohsenow (B&R) correlation for natural convection between vertical parallel plates gives an optimal fin spacing estimate:

$$S^* = 2.714 \left(\frac{H}{Ra_H^{1/4}} \right)$$

With $H = 37$ mm and $\Delta T = 55$ K: $Ra_H \approx 2 \times 10^5$, $S^* \approx 4.7$ mm, $N \approx 12$ fins. This grounded starting geometry was taken directly into Icepak.

2.4.2 Manual Sweep: Four Variables, No Map

The design space has four degrees of freedom: baseplate size, base thickness, fin count, fin height. A fin-count sweep revealed non-monotonic behaviour – $N = 9$ was better than $N = 12$ because narrower channels choke buoyancy flow (Table 5).

Table 5: Fin-count sweep at fixed geometry (12×12 cm, $t_b = 6$ mm, $\dot{Q} = 10$ W).

N	$T_{\text{chip,mean}}$ ($^{\circ}\text{C}$)	$T_{\text{chip,max}}$ ($^{\circ}\text{C}$)
3	98.8	117.9
6	83.8	102.8
9	77.9	96.6
12	82.8	101.6

After 17 manual runs and no clear convergence, the one-at-a-time approach was abandoned. Best manual result: $T_{\text{chip,max}} = 77.4\text{ }^{\circ}\text{C}$ with no confidence a better geometry didn't exist nearby.

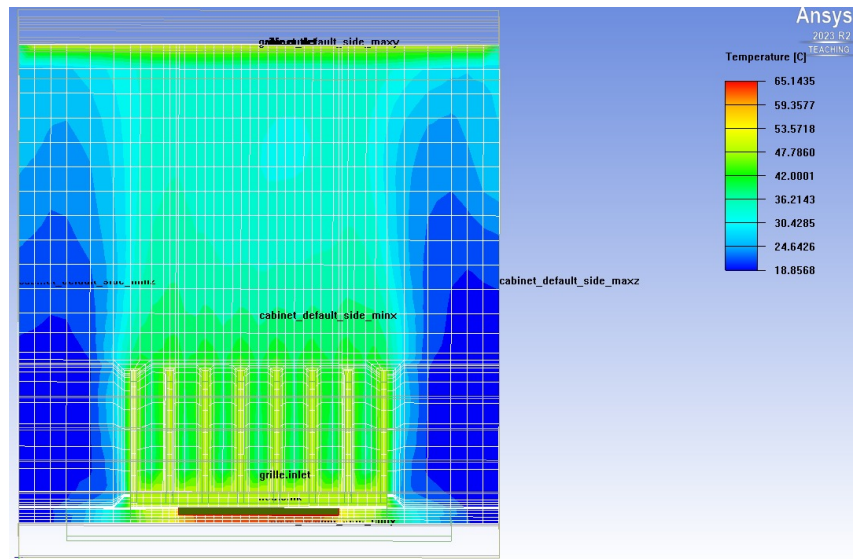


Figure 9: Icepak mesh used during the manual sweep phase. The mesh is regenerated for each candidate geometry.

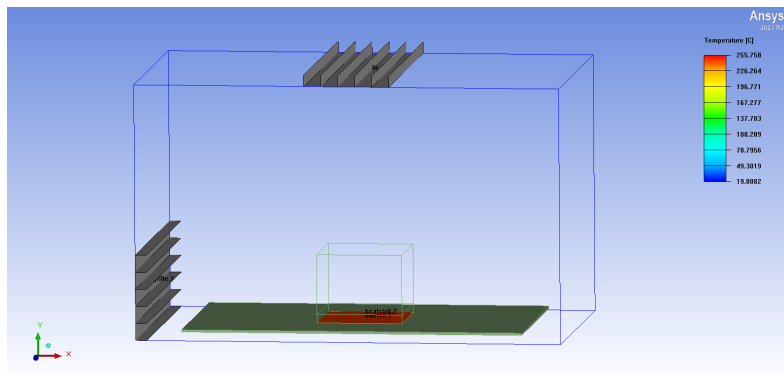


Figure 10: Temperature contour for an early 4-fin iteration. Fins are under-populated; hot spot persists at the chip base.

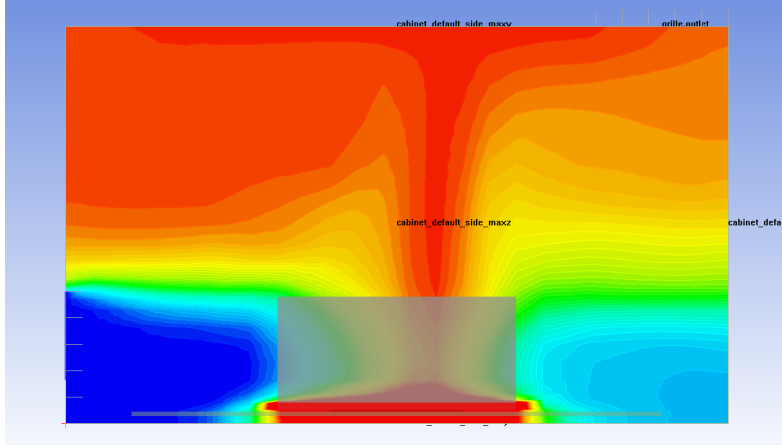


Figure 11: Buoyancy plume from iteration 2. Wider fin spacing allows stronger natural convection but leaves chip temperature above target.

2.4.3 Automated Optimisation

At this point the bottleneck was not the simulator – it was the strategy for choosing *which* designs to evaluate. With four coupled variables and a non-monotonic response, hand-picking the next trial kept opening new questions faster than it answered old ones. What was needed was a method that could learn from every completed run and direct the next batch toward regions of the parameter space that were either genuinely promising or genuinely unexplored.

A surrogate-model optimiser was therefore written in Python. It fits a cheap mathematical model to all results seen so far, predicts where the next batch should go, and improves its predictions with each new Icepak result.¹

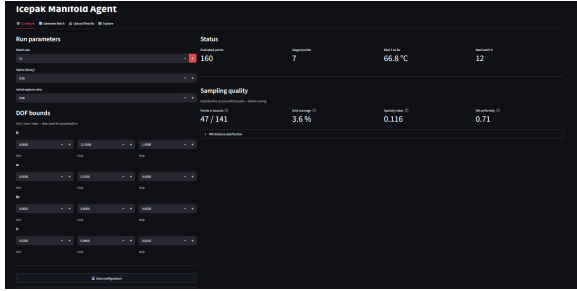


Figure 12: Surrogate generates the next candidate batch.

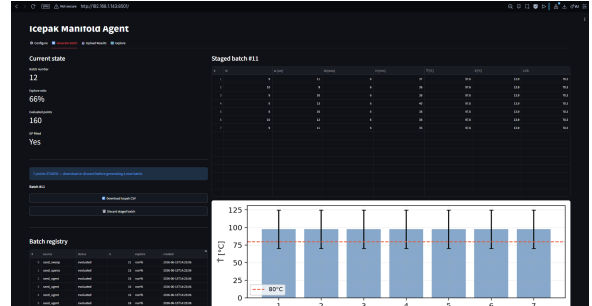


Figure 13: Batch preview before Icepak submission.

¹The surrogate combines a radial basis function (RBF) interpolator [2] with a Gaussian process regressor [3] fitted to the residuals. New candidates are chosen by minimising a lower confidence bound (LCB) acquisition function with an exponentially decaying exploration parameter κ [4]. Run history is stored in an SQLite database and managed through a Streamlit GUI. Full source code: <https://github.com/Efe0909/me324final>.

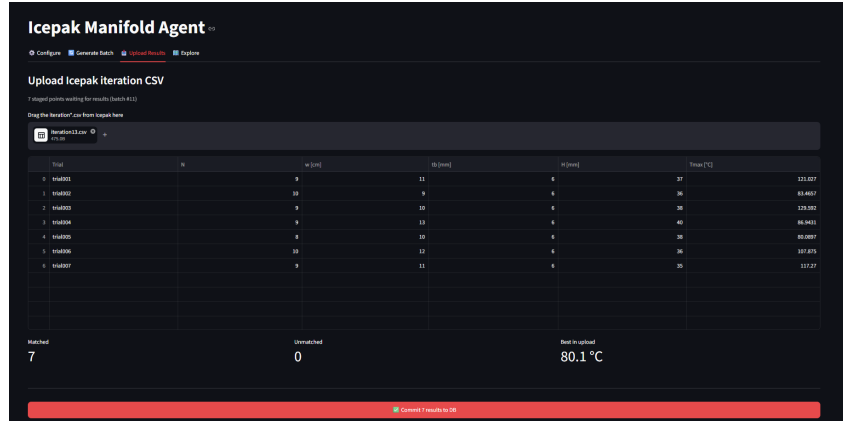


Figure 14: Upload tab: Icepak results imported back into the database after each batch.

For more information on surrogate optimisation methods, see [2], [4], and [3].

2.4.4 Optimisation Results

Six batches (~ 96 Icepak runs total). Figure 15 shows the surrogate response surface; Figure 16 shows convergence history. The flat tail after batch 3 confirms the search had adequately explored the space.

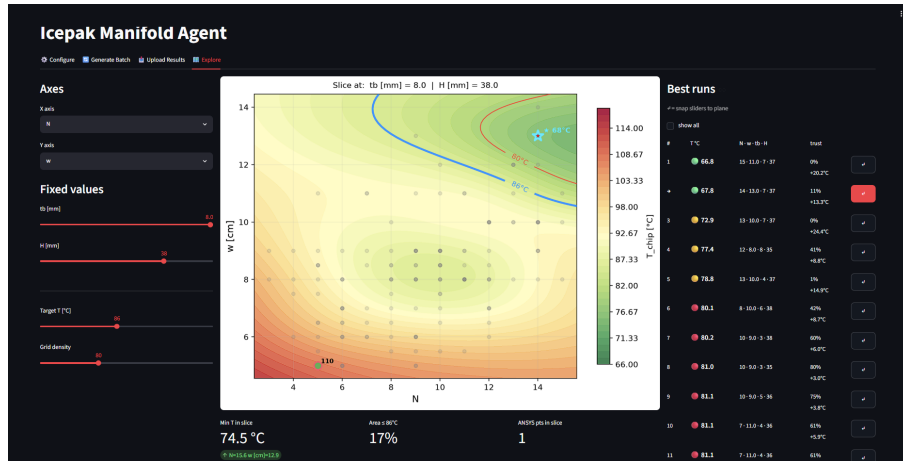


Figure 15: Surrogate response surface (RBF + GPR) sliced at fixed N and t_b . Grey dots are evaluated Icepak points; the blue contour marks the 80 °C target.

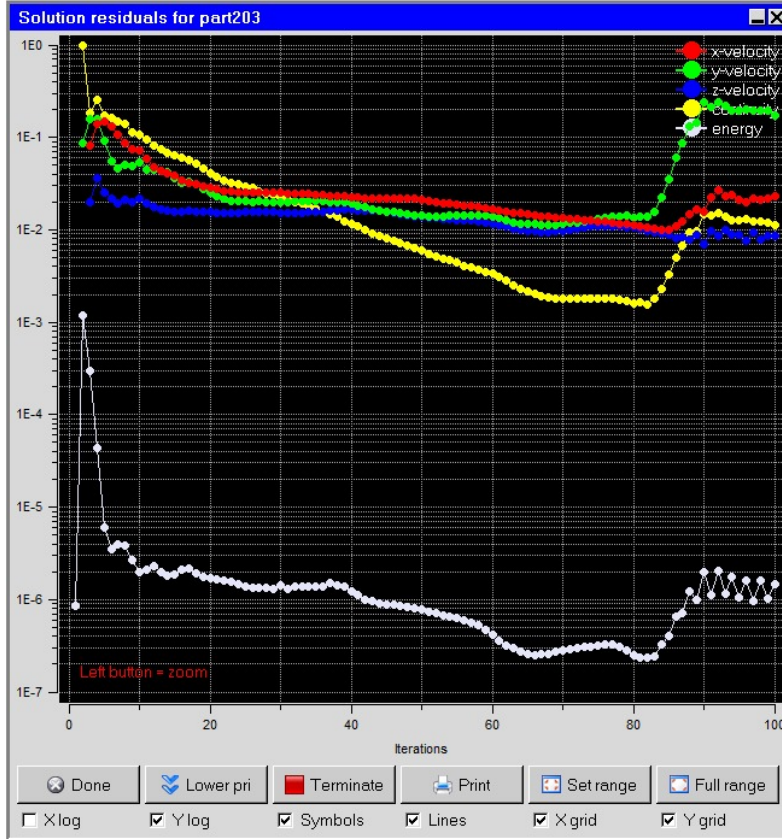


Figure 16: Convergence of the Bayesian optimisation: best-found $T_{\text{chip,max}}$ per batch.

2.5 Final Design and Results

2.5.1 Fin Thickness Refinement

After the Bayesian optimisation converged, a final one-variable sweep was conducted over fin thickness t_f while holding all other parameters fixed at the optimum values identified by the surrogate model. Three values were tested – 1.2, 1.5, and 2.0 mm – and the results are summarised in Table 6. Increasing fin thickness from 1.2 to 2.0 mm reduced the chip temperature by approximately 2.3 °C. The improvement is incremental rather than dramatic: thicker fins conduct heat more effectively along their height, but the added solid volume also narrows the inter-fin channel, partially counteracting the benefit. Nevertheless, the 2 mm fin was selected as the final geometry since the reduction comes at no cost in complexity.

Table 6: Effect of fin thickness on chip temperature at fixed optimum geometry ($N = 8$, $w = 8 \times 8$ cm, $t_b = 5$ mm, $H = 37$ mm).

Fin thickness t_f (mm)	$T_{\text{chip,max}}$ (°C)
1.2	67.6123
1.5	67.3679
2.0	65.2757

2.5.2 Final Design

The selected heat sink geometry is listed in Table 7. Figure 17 shows the Icepak model with all components labelled.

Table 7: Final heat sink design parameters.

Parameter	Symbol	Value
Baseplate footprint	$w \times w$	8×8 cm
Base thickness	t_b	5 mm
Number of fins	N	8
Fin height	H	37 mm
Fin thickness	t_f	2 mm
Heat sink volume	V	8.064×10^{-5} m ³
Material	–	Aluminium ($k = 202$ W/m·K)

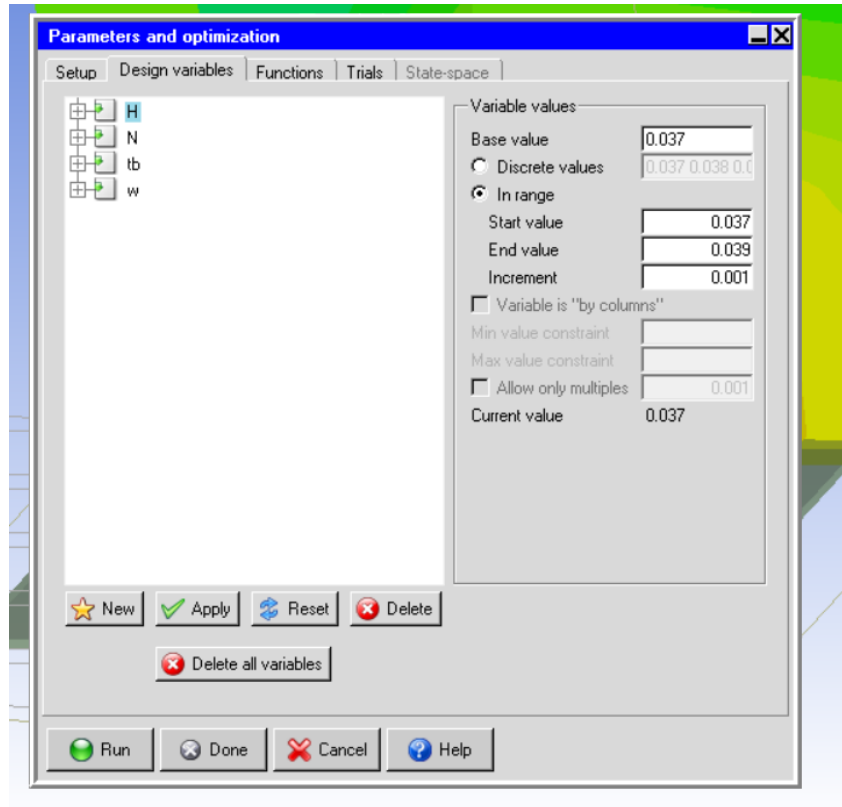


Figure 17: Final heat sink geometry as configured in Icepak.

This geometry is deliberately modest: with $N = 8$ fins on an 8×8 cm base the fin array does not approach the channel-choking regime identified in the fin-number sweep, and the footprint is only 60% larger than the chip itself. Larger bases and more fins do lower the chip temperature further, but the gains diminish rapidly beyond this point.

2.5.3 Thermal Performance

As a baseline reference, the chip was simulated without any heat sink: bare convection from the chip surface alone yielded $T_{\text{chip,max}} = 404.4\text{ }^{\circ}\text{C}$ and $T_{\text{chip,mean}} = 276.7\text{ }^{\circ}\text{C}$, confirming that natural convection directly from the chip is wholly inadequate for a 10 W source.

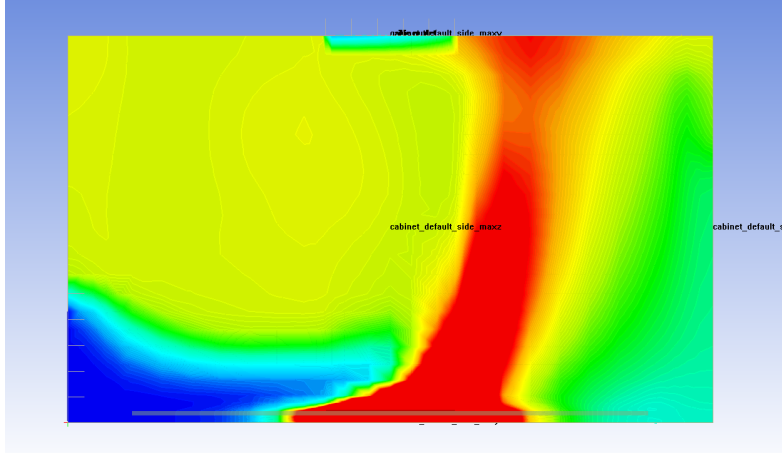


Figure 18: Temperature distribution (no heat sink). Chip temperature exceeds $400\text{ }^{\circ}\text{C}$.

With the final heat sink installed at $\dot{Q} = 10\text{ W}$, Icepak gives:

$$T_{\text{chip,max}} = 65.28\text{ }^{\circ}\text{C}, \quad R_{\text{total}} = \frac{65.2757 - 20}{10} = 4.528\text{ K/W}$$

Of this, the spreading resistance from the sub-model contributes less than 0.1 K/W ; the remaining 4.43 K/W is convective resistance of the fin array.

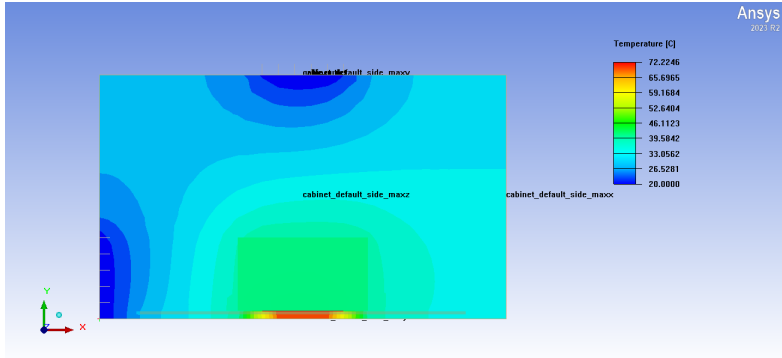


Figure 19: Temperature cross-section of the final design at $\dot{Q} = 10\text{ W}$. The chip hot spot is visible at the base (orange-red). The buoyancy-driven plume rises through the fin channels and exits at the top-centre outlet.

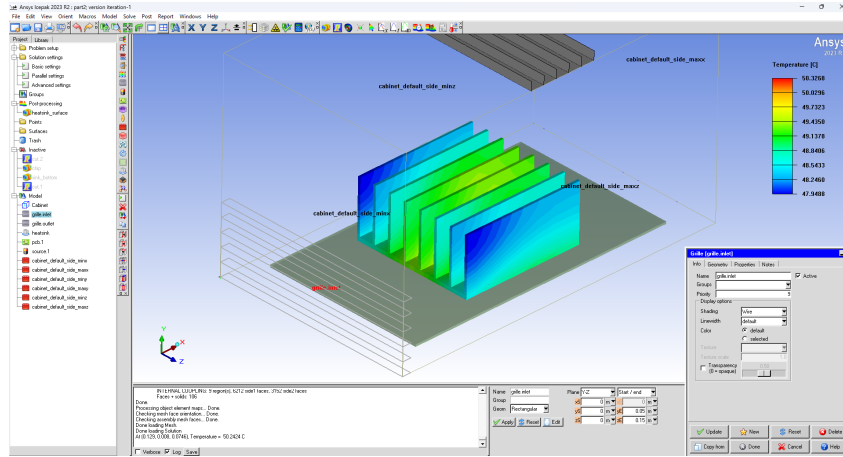


Figure 20: Temperature distribution from iteration 1 of the Bayesian optimisation. The surrogate had not yet identified the optimal channel spacing.

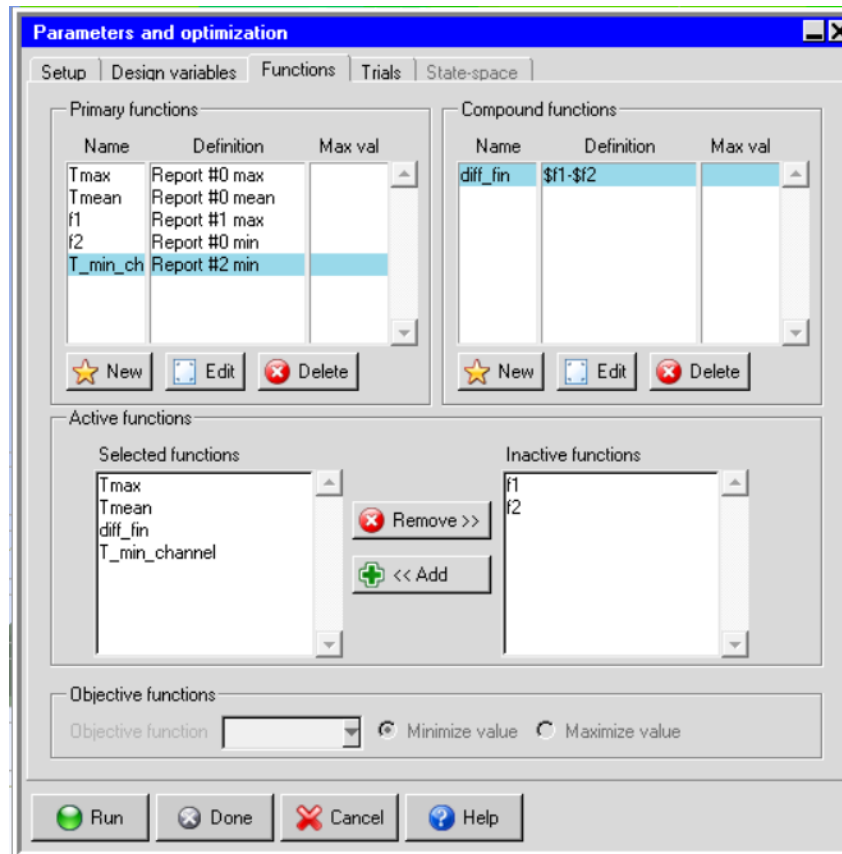


Figure 21: Optimisation target summary: $T_{\text{chip,max}}$ and R_{total} tracked across all evaluated designs.

2.5.4 Summary

Analytical B&R estimate → manual sweep (17 runs) → Bayesian optimisation (6 batches, ~96 runs) → fin-thickness refinement (3 runs). Final result: $T_{\text{chip,max}}$ reduced from 404.4 °C to **65.28 °C** (−339 °C) with a compact 8 × 8 cm aluminium array, natural convection only.

3 Part 3: Simscape Thermal Network

3.1 Objective

Part 3 uses the thermal resistance data produced by the Icepak heat sink simulation to build a lumped-parameter model of the chip–heat-sink–ambient system in MATLAB Simscape. The Icepak model was run at five power levels ($\dot{Q} = 2, 5, 10, 15, 20$ W) with the final heat sink geometry fixed, and the resulting chip temperatures were recorded. Because natural convection is a nonlinear process, the effective thermal resistance $R_{\text{eff}} = \Delta T / \dot{Q}$ varies with power; the Simscape model must capture this nonlinearity to be useful as a system-level tool.

3.2 Simscape Thermal Network

The Simscape model represents the thermal path from chip junction to ambient air as a network of thermal resistances and a heat source. The chip is modelled as an ideal heat source block dissipating $\dot{Q}$ W. The heat flows through the heat sink thermal resistance to a thermal ground set at $T_{\infty} = 20$ °C.

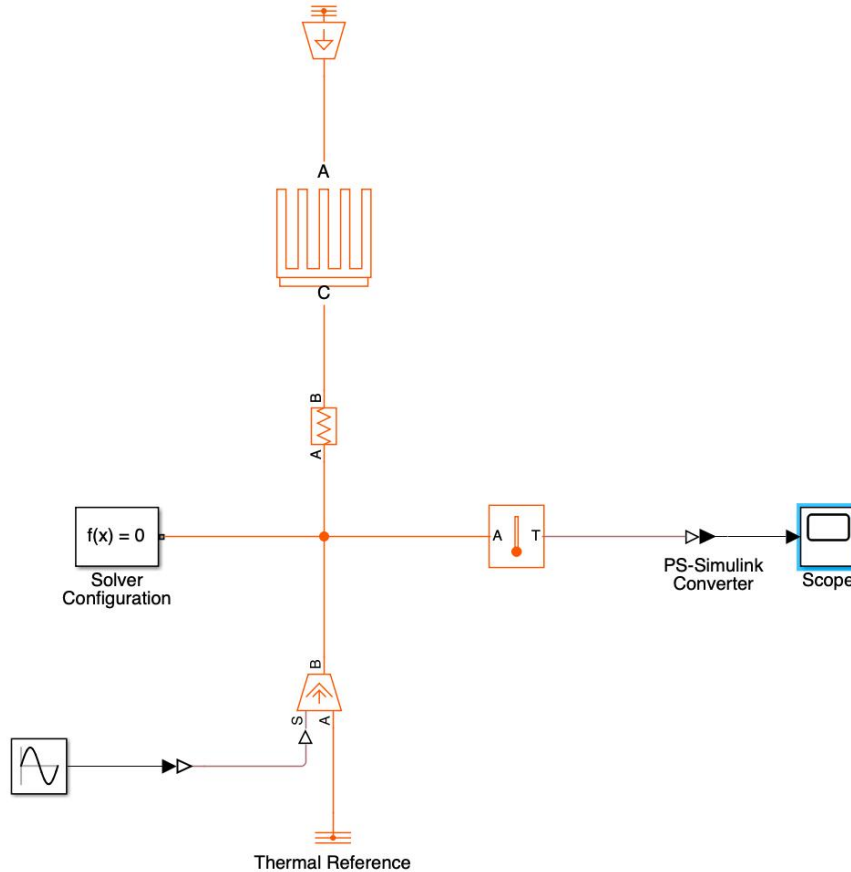


Figure 22: Simscape thermal network. The chip heat source drives heat through the heat sink resistance block to the ambient thermal ground.

The key challenge is that natural convection is nonlinear: the convective resistance de-

creases as the temperature difference and hence the buoyancy force increase. A simple constant resistance cannot reproduce the Icepak data across the full range of power levels. The thermal resistance block was therefore parameterised as a lookup table mapping $\dot{Q}$ to the effective resistance $R_{\text{eff}}(\dot{Q})$ derived from the Icepak sweep. Figure 23 shows the heat sink block parameters as configured.

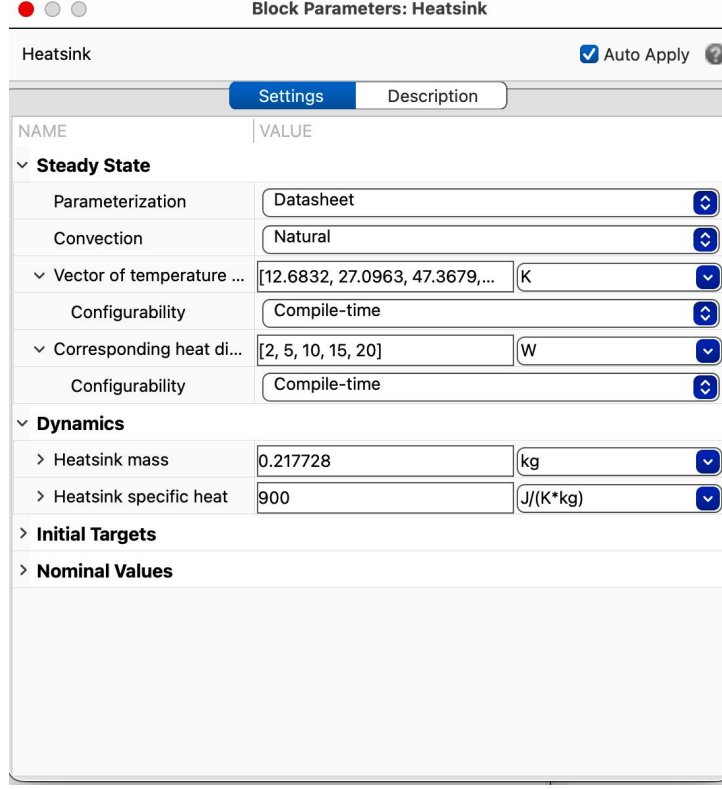


Figure 23: Heat sink thermal resistance block parameters in Simscape.

3.3 Results

3.3.1 Icepak $\dot{Q}$ - T Datasheet

Table 8 gives the chip temperature and effective thermal resistance from the Icepak sweep over heat dissipation levels. The ambient temperature is $T_{\infty} = 20$ °C in all cases.

Table 8: Icepak heat dissipation sweep with the final heat sink ($N = 8$, $w = 8 \times 8$ cm, $t_b = 5$ mm, $H = 37$ mm, $t_f = 2$ mm).

$\dot{Q}$ (W)	T_{chip} (°C)	$\Delta T = T_{\text{chip}} - T_{\infty}$ (K)	$R_{\text{eff}} = \Delta T / \dot{Q}$ (K/W)
2	32.6832	12.6832	6.34
5	47.0963	27.0963	5.42
10	67.3679	47.3679	4.74
15	105.011	85.011	5.67
20	117.253	97.253	4.86

R_{eff} is non-monotonic: 6.34 K/W at 2 W $\rightarrow$ 4.74 K/W at 10 W (stronger buoyancy as ΔT grows), then 5.67 K/W at 15 W (flow regime change). A single fixed resistance cannot reproduce this behaviour – the lookup table is necessary.

3.3.2 Simscape Output

Figures 25 and 26 show the heat flow and temperature histories from the Simscape simulation. The model reproduces the Icepak temperatures within the expected accuracy of the lookup-table interpolation.

Block Parameters: Sine Wave

Sine type determines the computational technique used. The parameters in the two types are related through:

Samples per period = $2\pi / (\text{Frequency} * \text{Sample time})$

Number of offset samples = $\text{Phase} * \text{Samples per period} / (2\pi)$

Use the sample-based sine type if numerical problems due to running for large times (e.g. overflow in absolute time) occur.

Parameters

Sine type: Time based

Time (t): Use simulation time

Amplitude: 2

Bias: 10

Frequency (rad/sec): $2\pi/60$ 0.10472

Phase (rad): 0

Sample time: 0

☒ Interpret vector parameters as 1-D

OK Cancel Help Apply

Figure 24: Sine wave block parameters used to drive the Simscape thermal network. Amplitude = 2 W, Bias = 10 W, frequency = $2\pi/60$ rad/s gives a power input oscillating between 8 and 12 W with a 60 s period.

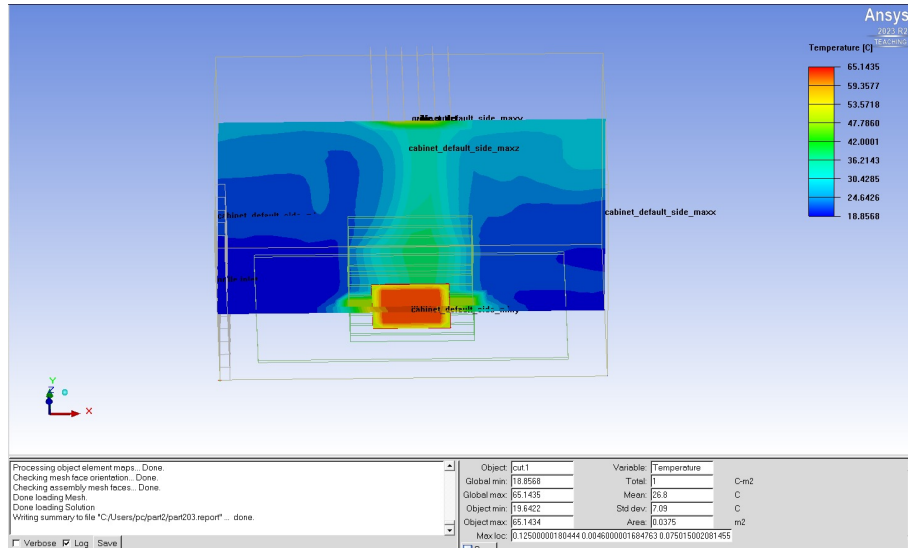


Figure 25: Simscape heat flow output for the chip-heat-sink-ambient network.

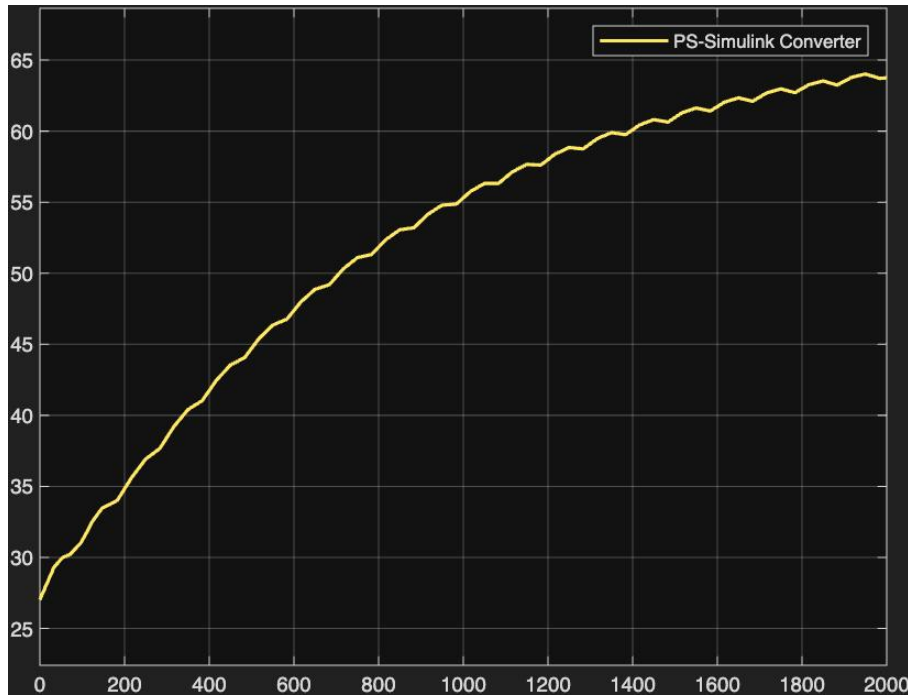


Figure 26: Simscape scope output showing chip temperature as a function of dissipated power.

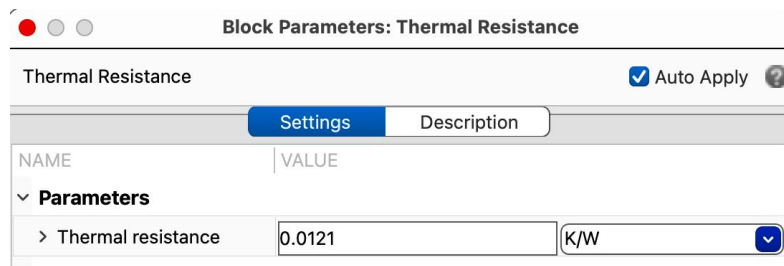


Figure 27: Effective thermal resistance R_{eff} versus $\dot{Q}$, used as the Simscape lookup table.

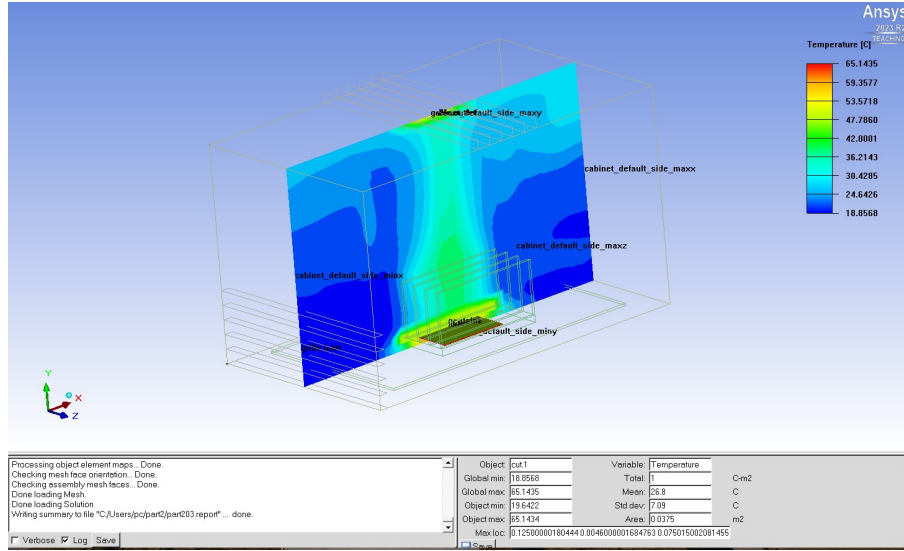


Figure 28: Simscape heat flow at intermediate power level.

References

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- [2] Donald R. Jones, Matthias Schonlau, and William J. Welch. Efficient global optimization of expensive black-box functions. *Journal of Global Optimization*, 13(4):455–492, 1998.
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- [4] Bobak Shahriari, Kevin Swersky, Ziyu Wang, Ryan P. Adams, and Nando de Freitas. Taking the human out of the loop: A review of Bayesian optimization. *Proceedings of the IEEE*, 104(1):148–175, 2016.